SOMAIYA VIDYAVIHAR

K J Somaiya Institute of Technology
(Formerly known as K J Somaiya Institute of Engineering and Information Technology)
An Autonomous Institute Permanently Affiliated to University of Mumbai.

Somaiya TRUST

S.Y. Mini Project Presentation

AgriAdvisor: An El Niño-Aware RAG-based Agricultural Advisory System

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PROJECT GUIDE

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INTRODUCTION

Background

- Domain: Applied AI for climate-resilient agriculture.
- El Niño can disrupt the Indian monsoon, causing rainfall deficits and crop stress across rainfed Maharashtra.
- AgriAdvisor combines verified government guidance with AI retrieval to produce localized, explainable advisories.

Problem Statement

- ICAR, IMD and climate-science sources publish reliable information, but it is fragmented across documents and portals.
- Extension workers need district-, crop- and scenario-specific guidance without manually searching multiple sources.
- General-purpose LLMs can be fluent without being grounded in the relevant official guidance.

CLIMATE SIGNAL

El Niño / ENSO

LOCAL CONTEXT

District • Crop • Month

ACTIONABLE ADVISORY

Recommendation → Reason → Source

OBJECTIVES

Primary Objective

Design and implement a Retrieval-Augmented Generation (RAG) system that synthesizes verified agricultural guidance into localized, explainable advisories for KVK officers and extension workers during El Niño scenarios.

Specific Goals

- Build a curated knowledge base using ICAR contingency plans, IMD advisories and verified ENSO literature.
- Retrieve guidance using district, crop, month and El Niño scenario as context.
- Generate traceable output in the format Recommendation → Reason → Source.
- Evaluate the grounded system against a plain LLM baseline.

LITERATURE SURVEY

Author(s) & Year	Paper Title	Method / Technique	Objective	Limitations	Key Findings	Research Gap
Soni, A.K., Tripathi, J.N., Sateesh, M. (2025)	Impact of soil moisture and rainfall variability on soybean crop yield during El Niño episodes in Maharashtra, India	GLDAS soil moisture + IMD rainfall; zonal statistics; Mann–Kendall trend test	Quantify El Niño effects on soil moisture, rainfall and soybean yield in Marathwada and Vidarbha	Sub-regional correlation differs; GLDAS resolution limits	Very strong El Niño: Marathwada −40% rainfall / −70% yield; Vidarbha −7% / −57%	Supports district-specific advisory logic rather than one uniform rule
Athira, K.S., Roxy, M.K., Dasgupta, P., Saranya, J.S., Singh, V.K., Attada, R. (2023)	Regional and temporal variability of Indian summer monsoon rainfall in relation to El Niño Southern Oscillation	PCA-Biplot analysis; 1901–2018 historical time series	Understand regional and temporal changes in ENSO–monsoon relationship	Coordinate boxes are not administrative districts	ENSO–monsoon relationship varies regionally and has weakened nationally since 1981	Justifies region-aware retrieval and geographic mapping

LITERATURE SURVEY

Author(s) & Year	Paper Title	Method / Technique	Objective	Limitations	Key Findings	Research Gap
Hill, S.A. et al. (2025)	More extreme Indian monsoon rainfall in El Niño summers	Observational rainfall analysis; 1901–2020	Determine El Niño effects on total and extreme daily monsoon rainfall	National-scale; not district-specific	El Niño lowers seasonal rainfall but can increase extreme daily rainfall risk	Advisories need separate deficit and extreme-rain warnings
Gurazada, M. et al. (2024)	El Niño and positive Indian Ocean Dipole conditions simultaneously reduce the production of multiple cereals across India	Climate indices + crop-production analysis	Assess combined ENSO/IOD effects on cereal production	Production-level analysis does not directly generate advisories	Combined climate anomalies can reduce multiple cereal yields	Motivates multi-source climate context in agricultural decisions

EXISTING SYSTEMS & RESEARCH GAP

EXISTING SYSTEM 1

RAG-based LLM Advisory Framework
Terna Engineering College, Navi Mumbai

- Uses Retrieval-Augmented Generation over Package of Practices documents for agricultural advisory.
- Covers multiple crops and compares different LLM approaches.

Limitation:
Does not explicitly use El Niño / ENSO scenario intensity as a retrieval-weighting variable.

EXISTING SYSTEM 2

SukhaRakshak AI
International Water Management Institute (IWMI)

- AI-based drought advisory approach involving drought indicators, contingency plans and multilingual delivery.
- Intended for farmers, extension officers and drought managers.

Limitation:
Focuses on general drought severity rather than explicit El Niño scenario-intensity retrieval.

RESEARCH GAP

Neither system makes El Niño scenario intensity an explicit retrieval-weighting variable.
AgriAdvisor combines verified government contingency plans, weather advisories and climate science with scenario-aware retrieval to generate district-, crop- and month-specific recommendations.

METHODOLOGY & ARCHITECTURE

Goal: Generate a localized, explainable agricultural advisory from verified sources.

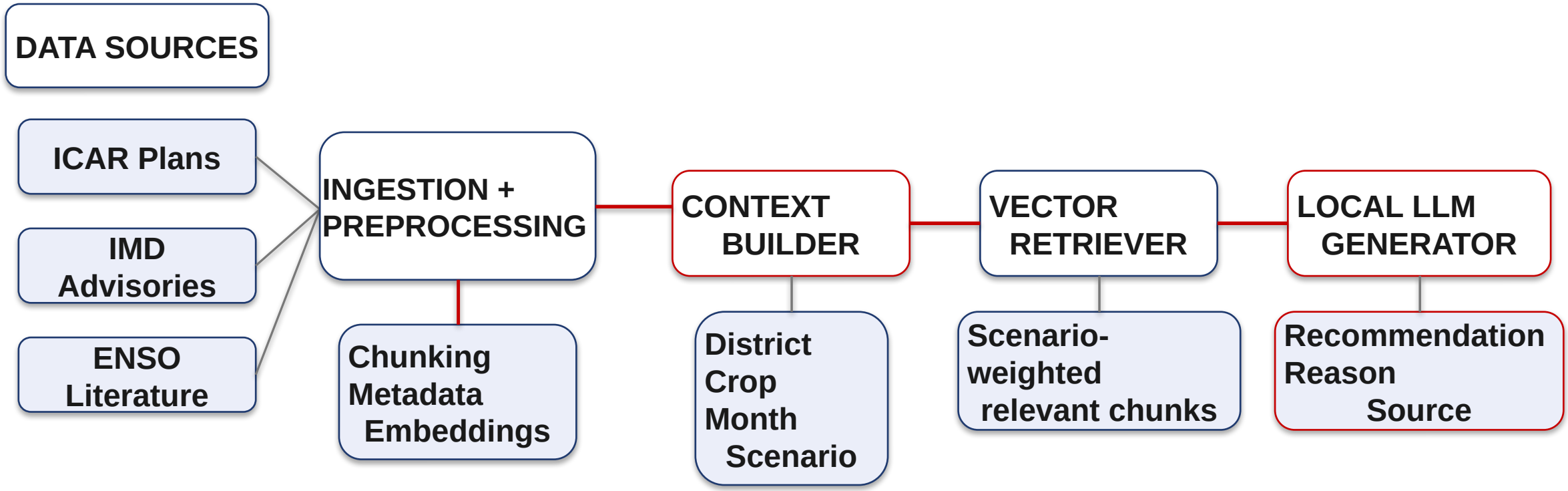
Approach

- Ingestion: ICAR contingency plans + IMD agromet advisories + ENSO literature.
- Preprocessing: semantic chunking, metadata tagging and embeddings.
- Retrieval: district + crop + month + scenario-aware search.
- Generation: local LLM produces Recommendation → Reason → Source.



The retrieval layer is the project's differentiator: the same question can surface different guidance when the climate scenario changes.

SYSTEM ARCHITECTURE



End-to-end traceability: every generated recommendation is anchored to retrieved source material.

TOOLS, TECHNOLOGIES & RESOURCES

Software & AI Stack

- Python — core implementation
- LangChain — retrieval orchestration
- Ollama — local LLM inference
- ChromaDB — vector database
- Streamlit — prototype interface

Knowledge Sources

- ICAR district contingency plans
- IMD Agromet Advisory Bulletins
- NOAA ONI / ENSO classification
- Peer-reviewed ENSO–India literature

Representative Districts

- Latur
- Beed
- Yavatmal
- Amravati

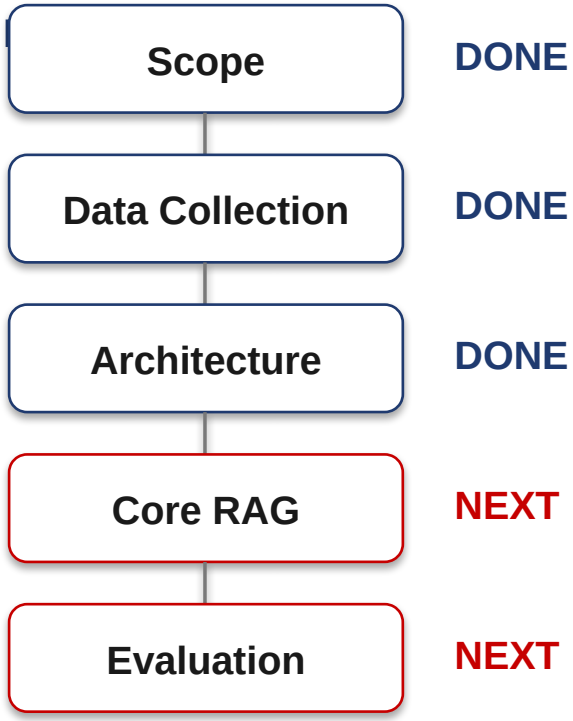
WORK COMPLETED SO FAR

Scope & Data

- Finalized problem scope, primary users, inputs and outputs.
- Selected four representative Maharashtra districts.
- Collected ICAR contingency plans and IMD advisory material.

Research & Design

- Built the requirements / knowledge-source matrix.
- Reviewed ENSO–monsoon and agricultural-impact literature.
- Defined scenario-aware retrieval and traceable output format.

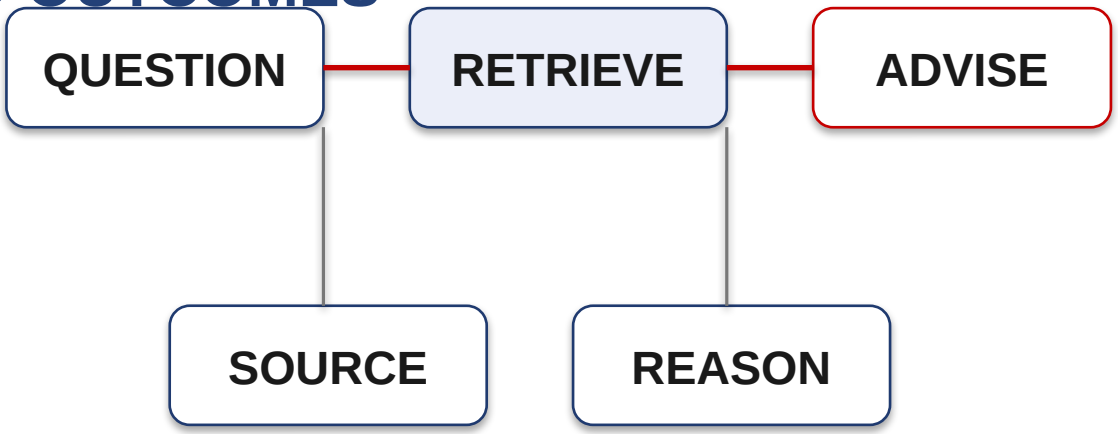


Current emphasis → implement the retrieval layer, connect the UI, and evaluate grounding against a plain LLM baseline.

FINAL / EXPECTED OUTCOMES

Working Prototype

- Streamlit-based advisory interface.
- District + crop + month + scenario inputs.
- Grounded response with Recommendation → Reason → Source.



Research Contribution

- Explicitly links El Niño scenario intensity to retrieval priority.
- Unifies government contingency plans, agromet advisories and climate literature.
- Creates a reproducible, explainable advisory pipeline.

Expected result: faster, more consistent and more defensible climate-aware agricultural guidance.

APPLICATIONS & FUTURE SCOPE

Applications

- KVK officers and agricultural extension workers
- District-level crop planning during El Niño years
- Pre-sowing advisories and contingency-plan retrieval
- Decision support for localized climate-risk communication

Future Scope

- Live IMD / weather feeds for dynamic context
- More districts, crops and regional languages
- Hybrid retrieval + reranking for stronger evidence selection
- Automated evaluation using citation correctness and groundedness
- Deployment for mobile / low-bandwidth extension workflows

TIMELINE

Activity	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
Literature Survey														
Synopsis Approval														
Requirement Gathering														
Dataset Collection														
System Design														
Core Implementation & Coding														
Mid-Term Review														
Testing & Debugging														
Result Analysis														
Final Report Writing														
Poster Preparation														
Final Viva														

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Thank You

Questions & Discussion

AgriAdvisor • Group 9

El Niño-aware • Grounded • Explainable